IASTAM 6.0 TECHNICAL CHALLENGE • TRACK 4 • PROBLEM 7

Transmitting Information, Not Raw Data

A satellite that sends what matters about ships, instead of sending pictures of empty sea

Project report

Team: <Your Team Name> — Computer Engineering, 2nd year

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1. What we were asked to solve

The IASTAM 6.0 challenge asks teams to work on the future of computing in space. We chose **Problem 7**, in the networks and communications track. Its statement is short:

> *"A satellite can generate a large volume of images while only a small part is genuinely useful to the mission. How can the volume to be transmitted be strongly reduced while preserving the information necessary for decision-making on the ground?"* The example given is a mission looking for **ships** at sea: should the satellite send all the images, or only the detections and the zones of interest?

The jury evaluates four things: how much the data volume is reduced, whether the useful information survives, how accurate the results are, and how long the ground waits for them.

2. Why this is a real problem

- A satellite can only send data while it passes over a ground station. The rest of the time the radio is useless, and the images pile up onboard.
- Most of what a camera records at sea is worthless: empty water, or clouds hiding everything.
- Sensors keep improving, so satellites produce more data every year, while the radio link does not improve at the same speed.

We did not take these statements on trust. We computed the real orbit of a satellite at 500 km above the Earth and the visibility of a ground station in **Sfax, Tunisia**:

What we computed	Result
Passes over the station per day	5
Total contact time per day	34.2 minutes
Longest wait between two contacts	11.4 hours
Data one pass can carry (small satellite radio)	130 MB on average

So the satellite is alone with its data for hours, and when the radio finally works it has a few minutes. Sending raw pictures is hopeless: a single large image can fill an entire pass.

3. What is done today

Approach	What it does	Its limit
Send everything ("bent pipe")	The satellite is a simple relay	The link saturates; most of what arrives is empty sea
Compress the images	Standard image compression	A smaller picture of empty water is still a picture of empty water
Throw away bad images (ESA Φ-sat-1, 2020)	A neural network onboard rejects cloudy images	Only decides keep or discard
Send detections (ESA Φ-sat-2, 2024)	Detects ships onboard, sends a small patch and a position	Sends the same amount for every ship, whether obvious or doubtful

Nobody, in what we read, decides **how much detail each detection deserves** and **in which order to send things** when several hours of data are waiting for a few minutes of radio. That is the gap we work on.

4. Our solution

The satellite stops behaving like a camera that ships pictures, and behaves like an observer that reports what it sees. Two decisions are ours: **how many bytes each ship deserves**, and **what to send first**.

(figure missing: simple_steps.png)

Decision 1 — how much detail per ship

A detector gives each ship a confidence score. Most systems send less when they are unsure. We do the opposite: **the less certain the satellite is, the more evidence it sends**.

Situation	What is sent	Size
Certain, and the ship already announces itself by radio (AIS)	Just the facts: position, size, heading	about 40 bytes
Uncertain	A small image of the ship so the ground can confirm	1–3 KB
A ship with no radio identification (“dark vessel”)	A larger image that can be cut short if the radio time runs out	10–50 KB
A port, or a coastline full of boats	The compressed area itself, because detectors are unreliable there	10–80 KB
Nothing convincing found	A tiny preview of the whole tile, as a safety net	about 1 KB

Decision 2 — what to send first

Everything waiting onboard is queued with a value and a size. When a pass starts, the satellite sends the items with the best **value per byte** first, items that have waited a long time gain priority, and large images can be cut short when the pass ends. This is the classic knapsack problem: fill a limited bag with the most valuable objects.

A missed ship is never silently lost: a preview of every tile is sent, the raw data stays onboard for a while so the ground can ask for it again, and the ground checks detections against public ship-tracking data.

5. What we built and what it shows

Everything below runs today as a Python program, with 28 automatic tests. Two kinds of results must not be confused: measurements on **real satellite data**, and a **simulation** of a day in orbit where the ships are generated by us.

Real data: cleaning the ship dataset

We use the Airbus Ship Detection dataset: real satellite images at 1.5 m per pixel, cut into tiles of 768 × 768 pixels. We kept 53,195 tiles containing 81,723 ships.

These tiles are cut out of larger scenes, and neighbouring cuts overlap, so **the same ship can appear in two different images**. If one copy trains the detector and the other tests it, the test is meaningless. We therefore built a tool that proves when two tiles show the same place, and keeps such tiles on the same side of the split.

Measured on the real dataset	Result
Tiles analysed	53,195
Pairs of tiles proven to show the same place	6,282
Tiles having at least one duplicate	10,141 (19.1%)
Groups a naive random split would have leaked	1,617
Groups leaking in our split	0

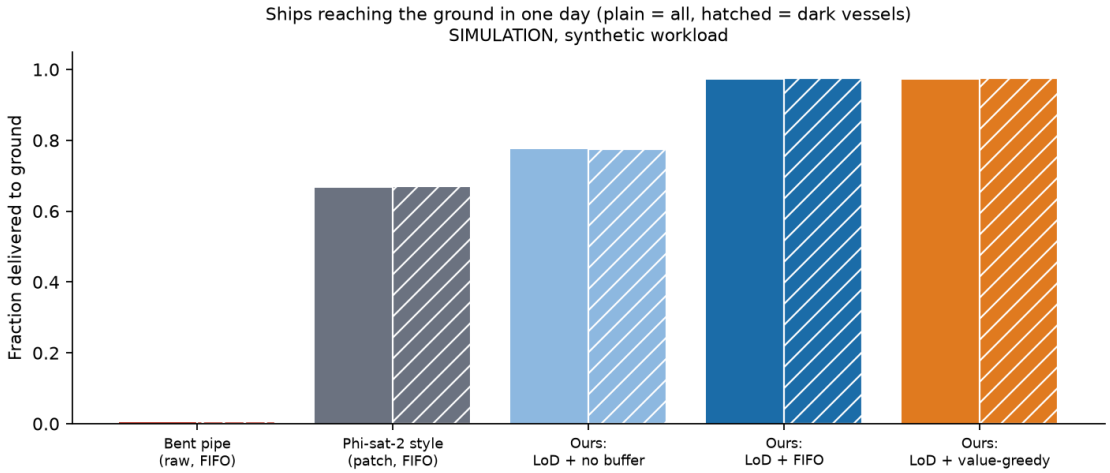
Simulation: one day in orbit

We simulate a full day: the real passes over Sfax, a small-satellite radio, and a stream of images with ships. We then compare what reaches the ground under different strategies.

Strategy	Ships that reach the ground	Of which unidentified ships
Send raw images	0.4%	0.6%
Send a fixed patch per ship (Φ-sat-2 style)	66.6%	67.1%
Our detail levels, oldest sent first	97.1%	97.6%
Our detail levels + our priority order	**97.1%**	**97.6%**

Our satellite offers **136 MB** instead of 60 GB of raw images — about **442 times less data** — and still reports 97.1% of the ships. The payload sizes used here were measured on real Airbus ship images, not guessed.

At this imaging rate everything fits in the available radio time, so the sending order changes the **waiting time** rather than the number of ships: an unidentified ship arrives after 4.2 h instead of 5.4 h. The order becomes decisive when the satellite images more: at twice this rate the simple order delivers 69% of ships and ours 97%; at four times, 34% against 83%.



Simulation: ships reaching the ground in one day. Plain bars: all ships. Hatched bars: ships without radio identification.

6. What is not finished, and what is next

Question	Where we stand
Does the detector work on real images?	The dataset is clean and ready; training is the next step
Is the confidence score trustworthy?	Detectors are usually over-confident; we will correct it and publish the check
Are the byte sizes realistic?	Today they are estimates; we will measure them on real ship images
Is the sending order the best possible?	We will compare it with the exact optimum on small cases
What about ships cut between two tiles?	Large ships are cut 53–84% of the time; we propose to re-examine only the borders, at about 1% extra computing
Does it work on other satellites?	Planned test on free Sentinel-2 images of the Tunisian coast

Honesty rule we follow

- Every figure says whether it is a measurement on real data, a simulation, or a number from a published paper.
- We list the weaknesses of our own work, including a comparison that currently favours us unfairly.
- Three versions of our duplicate-finding tool were thrown away after we looked at the evidence and saw it was wrong: unrelated images of empty sea, of fog, and of scene edges were being matched.

7. How to present this in three minutes

Time	What to say
0:00–0:30	A satellite sees far more than it can send: five passes a day over Sfax, 34 minutes of radio, up to eleven hours of silence. And most images show empty sea.
0:30–1:00	So we stop sending pictures. The satellite finds the ships and reports them. Show the six-step figure.
1:00–1:45	Our first idea: the less sure the satellite is, the more evidence it sends. A certain, identified ship costs forty bytes; a suspicious one gets a real image.
1:45–2:30	Our second idea: when the radio opens, send the most valuable bytes first. In our simulation this delivers 97% of the ships with about 440 times less data than raw images, and suspicious ships arrive an hour earlier — much more when the satellite images heavily.
2:30–3:00	What is real and what is not: the orbit and the dataset cleaning are measured, the day in orbit is simulated, and the detector is our next step.

Expect these questions: what if a ship is missed (preview + data kept onboard + the ground can ask again), how do you know your test is not inflated (we proved and removed the duplicates), and does “no radio identification” prove wrongdoing (no — it raises priority, a human decides).